

Task 1 starting nodes:

The screenshot displays a ROS2 environment with the following components:

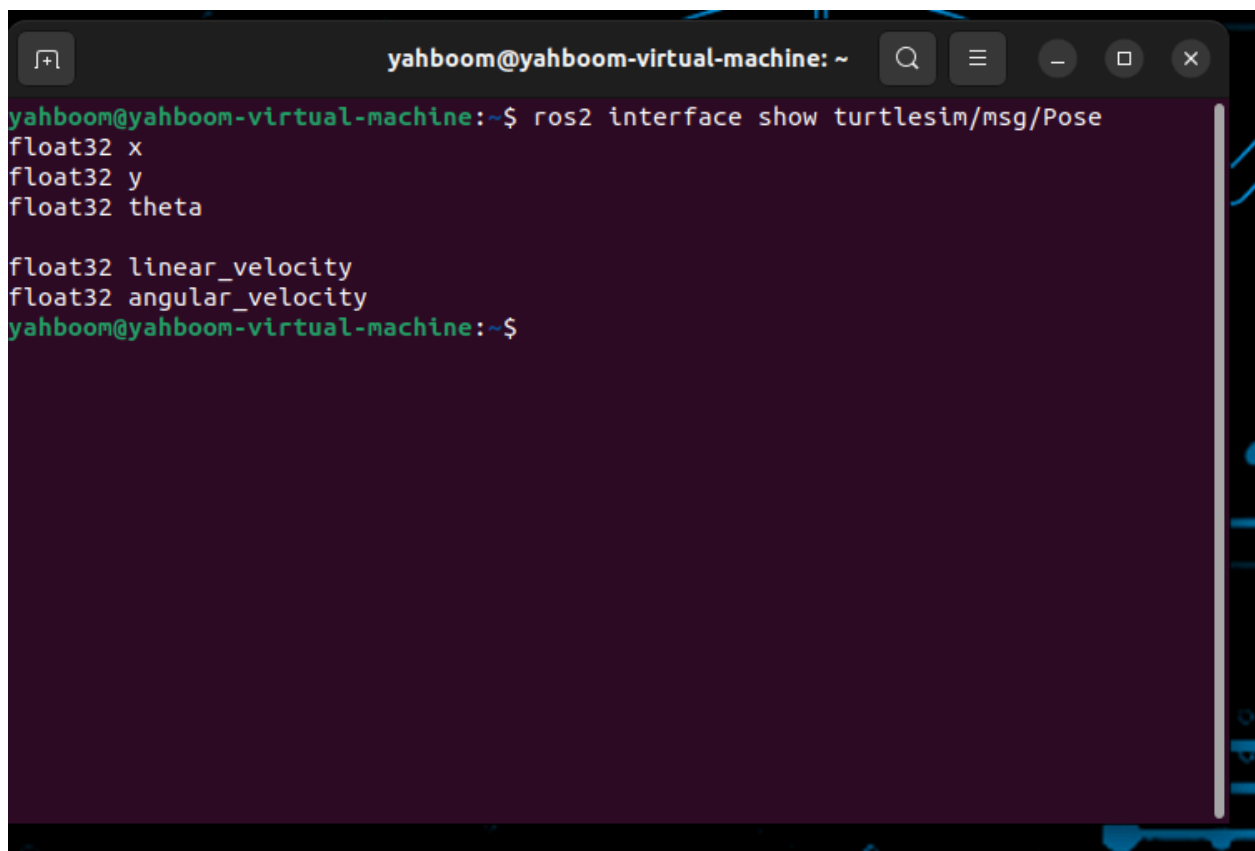
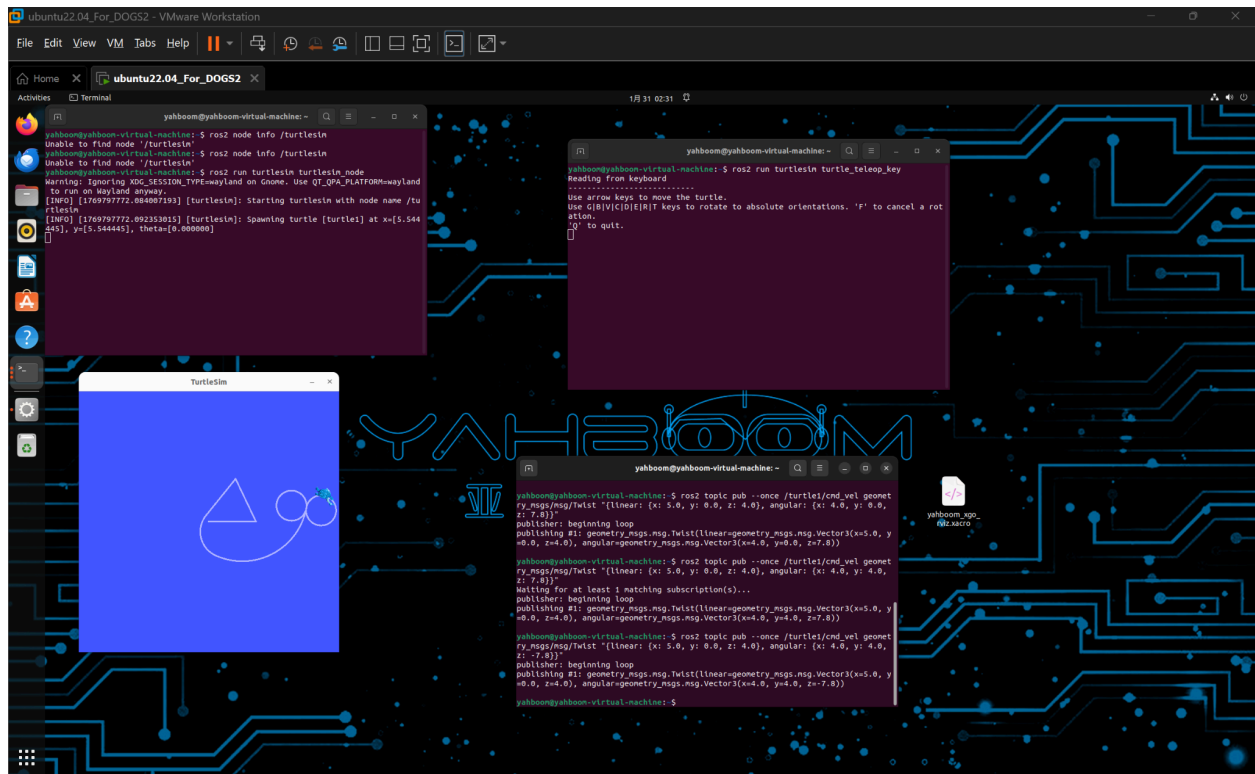
- Terminal Window (Top Left):** Shows the output of the `rostopic echo /turtle1/pose` command, displaying the turtle's position and orientation. The output is as follows:

```
[x=1.120785, y=5.818495]
[1769795851.857878912] [turtle1]: Oh no! I hit the wall! (Clamping from
[x=1.120785, y=5.818495]
[1769795851.874818521] [turtle1]: Oh no! I hit the wall! (Clamping from
[x=1.120785, y=5.818495]
[1769795851.890804844] [turtle1]: Oh no! I hit the wall! (Clamping from
[x=1.120785, y=5.818495]
[1769795851.90866439] [turtle1]: Oh no! I hit the wall! (Clamping from
[x=1.120785, y=5.818495]
[1769795851.926473188] [turtle1]: Oh no! I hit the wall! (Clamping from
[x=1.120785, y=5.818495]
[1769795851.94428197] [turtle1]: Oh no! I hit the wall! (Clamping from
[x=1.120785, y=5.818495]
[1769795851.96209076] [turtle1]: Oh no! I hit the wall! (Clamping from
[x=1.120785, y=5.818495]
[1769795851.97989955] [turtle1]: Oh no! I hit the wall! (Clamping from
[x=1.120785, y=5.818495]
[1769795851.99770834] [turtle1]: Oh no! I hit the wall! (Clamping from
[x=1.120785, y=5.818495]
```
- Terminal Window (Top Right):** Shows the output of the `rostopic echo /turtle1/pose` command, displaying the turtle's position and orientation. The output is as follows:

```
angular:
x: 0.0
y: 0.0
z: 0.0
...
linear:
x: 0.0
y: 0.0
z: 0.0
angular:
x: 0.0
y: 0.0
z: 0.0
...
linear:
x: 2.0
y: 0.0
z: 0.0
angular:
x: 0.0
y: 0.0
z: 0.0
...
^Cyahboom@yahboom-virtual-machine: $
```
- TurtleSim Window (Bottom Left):** A window showing a turtle in a 2D environment, with a red line indicating its path.
- Terminal Window (Bottom Right):** Shows the output of the `rostopic echo /turtle1/pose` command, displaying the turtle's position and orientation. The output is as follows:

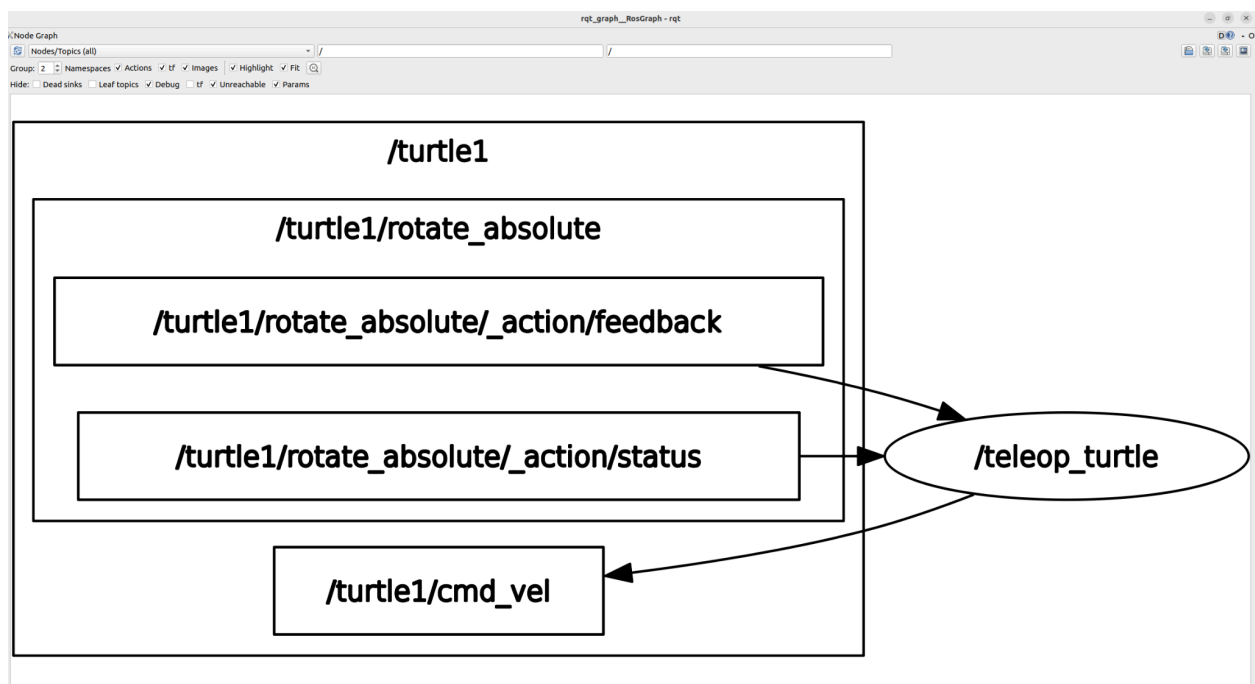
```
angular:
x: 0.0
y: 0.0
z: 0.0
...
linear:
x: 0.0
y: 0.0
z: 0.0
angular:
x: 0.0
y: 0.0
z: 0.0
...
linear:
x: 2.0
y: 0.0
z: 0.0
angular:
x: 0.0
y: 0.0
z: 0.0
...
^Cyahboom@yahboom-virtual-machine: $
```
- Node Graph Window (Bottom Right):** A window showing the ROS2 node graph. The graph includes the following nodes and topics:
 - Nodes:** `/turtle1/absolute`, `/turtle1/absolute/actionfeedback`, `/turtle1/absolute/actionstatus`, `/turtle1/vel`.
 - Topics:** `/turtle1/pose`, `/turtle1/pose`, `/turtle1/pose`.

Steps 7,21,25

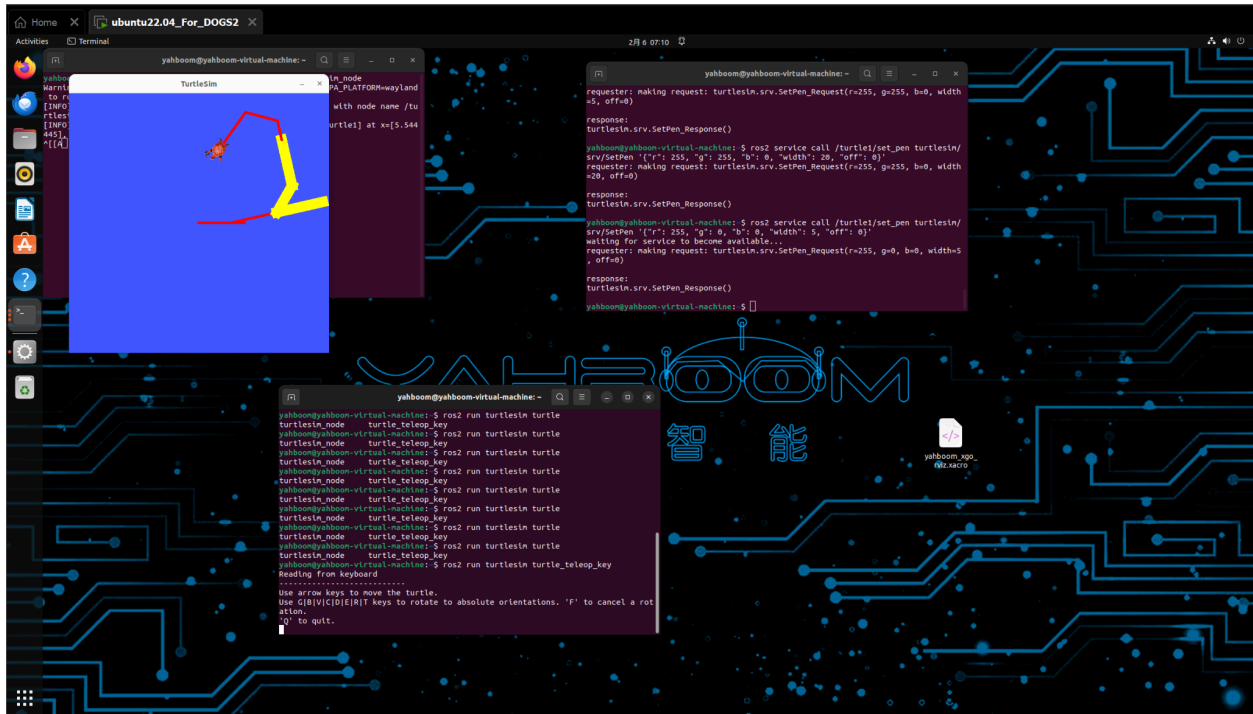


```
yahboom@yahboom-virtual-machine: ~  
angular_velocity: 0.0  
---  
x: 7.869671821594238  
y: 3.176300287246704  
theta: 0.4259234070777893  
linear_velocity: 0.0  
angular_velocity: 0.0  
---  
x: 7.869671821594238  
y: 3.176300287246704  
theta: 0.4259234070777893  
linear_velocity: 0.0  
angular_velocity: 0.0  
---  
^Z  
[1]+  Stopped                  ros2 topic echo /turtle1/pose  
yahboom@yahboom-virtual-machine:~$ ^C  
yahboom@yahboom-virtual-machine:~$ ros2 interface show turtlesim/turtle1/pose  
Could not find the interface '/opt/ros/humble/share/turtlesim/turtle1/pose.idl'  
yahboom@yahboom-virtual-machine:~$ ros2 topic info /turtle1/pose  
Type: turtlesim/msg/Pose  
Publisher count: 1  
Subscription count: 0  
yahboom@yahboom-virtual-machine:~$
```

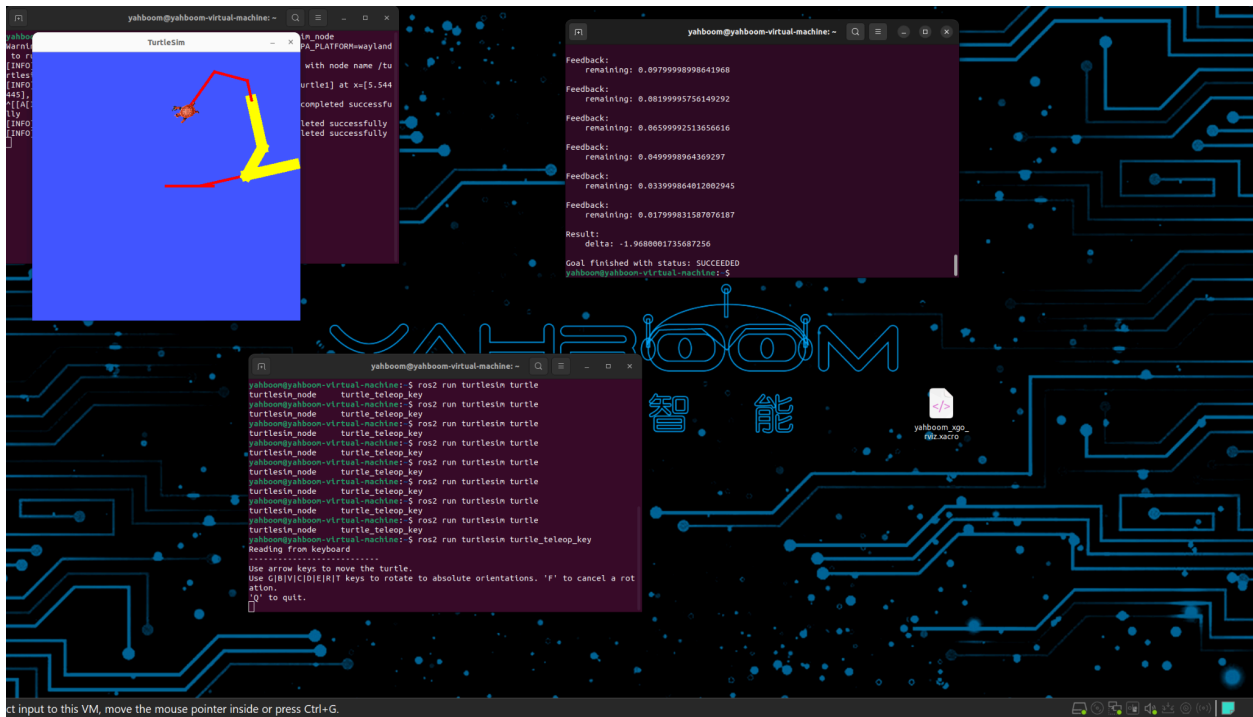
```
yahboom@yahboom-virtual-machine: ~  
linear_velocity: 0.0  
angular_velocity: 0.0  
---  
x: 7.869671821594238  
y: 3.176300287246704  
theta: 0.4259234070777893  
linear_velocity: 0.0  
angular_velocity: 0.0  
---  
x: 7.869671821594238  
y: 3.176300287246704  
theta: 0.4259234070777893  
linear_velocity: 0.0  
angular_velocity: 0.0  
---  
x: 7.869671821594238  
y: 3.176300287246704  
theta: 0.4259234070777893  
linear_velocity: 0.0  
angular_velocity: 0.0  
---  
^Z  
[1]+  Stopped                  ros2 topic echo /turtle1/pose  
yahboom@yahboom-virtual-machine:~$
```



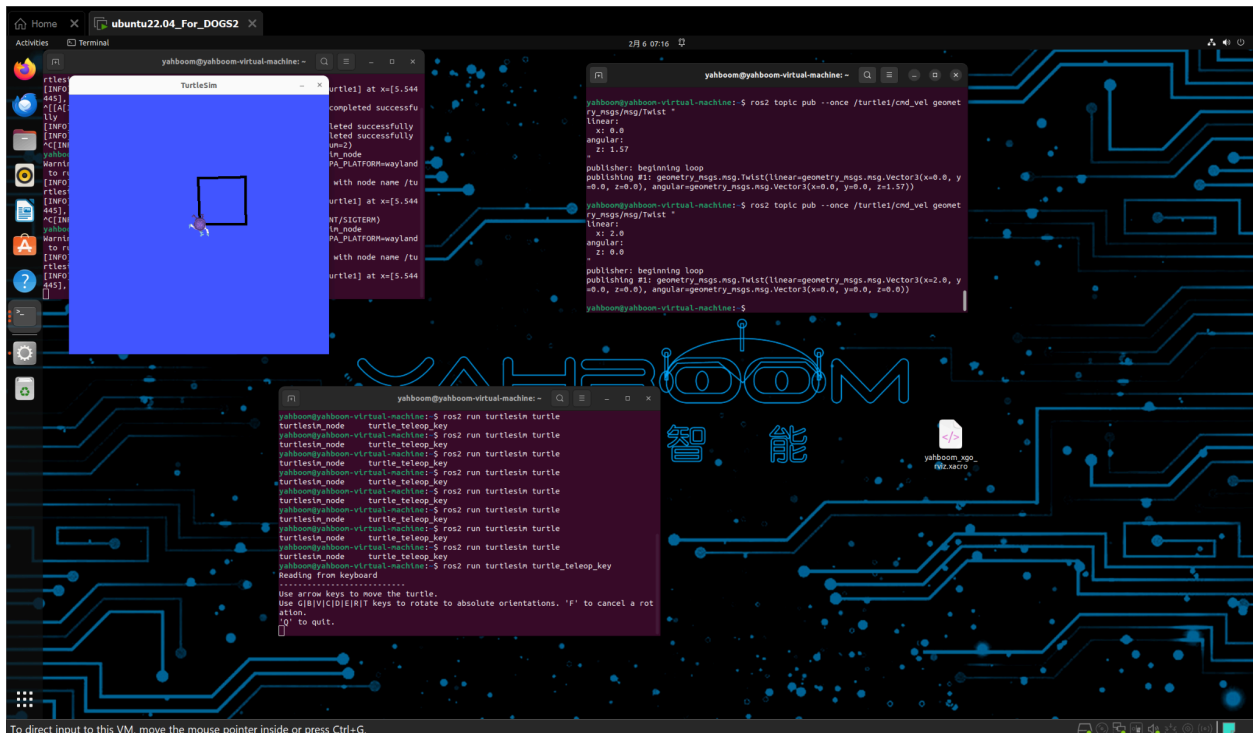
Challenge:



Task 8:



Task 3:



First I used the command from the challenge to change the pen to black, then used the commands “ros2 topic pub --once /turtle1/cmd_vel geometry_msgs/msg/Twist ”

linear:

x: 2.0

angular:

z: 0.0

”

“ and “ros2 topic pub --once /turtle1/cmd_vel geometry_msgs/msg/Twist ”

linear:

x: 0.0

angular:

z: 1.57

”

“ to move the turtle forward then turn 90 degrees then repeat till the square was done.